

MOLECULAR DISTRIBUTION IN LIPID NANOPARTICLES IN THE PRESENCE OF DIFFERENT mRNA TOPOLOGIES AND IONIZABLE LIPIDS

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ABSTRACT

Lipid nanoparticles (LNPs) are the leading delivery vehicles for mRNA therapeutics, including vaccines. Despite their clinical success, key challenges persist regarding mRNA stability, encapsulation efficiency, and endosomal release. The molecular-level understanding of how RNA topology influences the internal organization of LNPs remains limited. In particular, the formation of water pockets and bleb-like structures within LNPs is poorly characterized, yet these features may critically affect mRNA delivery. Molecular dynamics simulations with Martini coarse-grained (CG) models allowed modeling the complexity of LNPs and studying their internal organization (1,2). Therefore, in this work, we use Martini CG simulations to study how different RNA topologies and ionizable lipids (ILs) affect molecular distribution in LNP formulations. We established a minimal simulation setup based on Moderna and Pfizer/BioNTech LNP compositions, incorporating the clinically relevant ILs MC3, SM-102, or ALC-0315 with DSPC helper lipids and cholesterol. System parameters were optimized to balance computational tractability with sensitivity to RNA-induced organizational changes. We found that mRNA alters IL distribution and water pocket formation in an IL-dependent manner. This framework enables systematic characterization of RNA topology effects on LNP organization and provides a basis for evaluating newly designed ILs toward more effective mRNA delivery systems.

REFERENCES

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